

Name : _____ R. No. _____ Class/ Sec: _____ Date: _____ Invig. Sign _____

ATOMIC ENERGY CENTRAL SCHOOL, NARORA
CLASS IX MATHS UNIT TEST FIRST 2018-19

MM : 40

THE LIFE

General Instructions:

- All questions are compulsory and marks are mentioned in front of each question.
- Electronic devices are prohibited to use in the examination.
- Use Blue or Black Pen only.

SECTION-A

$$1 \times 3 = 3$$

1. Express $\frac{34}{9}$ in the decimal form.
2. Write the co-efficients of x^2 in $2 - x^2 + x^3$.
3. Find the co-ordinate of the point which lies on x and y axes both.

SECTION-B

$$2 \times 3 = 6$$

4. Find two rational numbers between $\frac{1}{2}$ and $\frac{3}{4}$.
5. Factorize: $16x^4 - 1$.
6. Verify that -1 is the zero of the polynomial $p(x) = 2x^3 - 9x^2 + x + 12$.

SECTION-C

$$3 \times 5 = 15$$

7. Show that $(x + 3)$ is a factor of $p(x) = 69 + 11x - x^2 + x^3$.
8. In which quadrant do the following points lie?
a) $(-6, 5)$ b) $(2, -9)$ c) $(3, 4)$
9. A point lies on the x -axis at a distance of 7 units from the y -axis. What are its co-ordinates? What will be the co-ordinate if it lies on y -axis at a distance of -7 units from x -axis?
10. Find three different solutions for the equation: $3x - 8y = 27$.
11. Find the value of k if $x = 3$ and $y = 1$ is solution of the equation $x + 5y = k$.

SECTION-D

$$4 \times 4 = 16$$

12. If $a + b = 9$ and $ab = 7$, find the value of $a^3 + b^3$.
13. Rationalise the denominator of $\frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} - \sqrt{3}}$.
14. Plot the points A(0,4), B(-3, 0) and C(2, 0) on a graph paper. What figure will we get on joining these points? Find the area of the figure.
15. Give the geometrical representation of $x = -3$ as an equation in two variables.